

GOLDEN ARROWS DATASET V3.0

Simulation-Generated Multi-Drone Swarm
Transition Dataset

Dataset Description

Version 3.0

100 Missions

11,570 Transitions

347,100 Drone-Transition Records

30 Drones per Transition

A technical description of the frozen simulation-generated V3.0 release. No authorship, affiliation, DOI, URL, or performance claim is asserted here because those metadata are not finalized in the package.

Contents

Contents	2
1. Abstract	3
2. Dataset Identity and Motivation	3
3. Dataset Composition	3
4. Package Organization	4
5. Simulation and Mission Context	5
6. Data Generation and Reproducibility	5
7. Conceptual System Architecture	5
8. Transition Structure	5
9. Advanced Swarm Behavior	5
10. State-Only Preprocessing Example	7
11. Behavior-Aware Preprocessing Example	7
12. Temporal Semantics	8
13. Data Leakage and Feature Interpretation	9
14. Mission-Level Splitting	9
15. Running the Optional Examples	9
16. Validation and Quality Control	9
17. Validation Artifacts	9
18. Intended Research Use	10
19. Limitations	10
20. Publication Scope	10
21. Final Package Structure	11
22. Related Documentation and Citation	11
23. Closing Statement	11

1. Abstract

Golden Arrows Dataset V3.0 is a simulation-generated multi-drone swarm transition dataset containing 100 missions, 11,570 rich transitions, and 347,100 drone-transition records from a 30-drone swarm. The release covers PATROL, SURVEILLANCE, SEARCH, and RECOVERY mission types and preserves mission context, decisions, actions, results, network state, drone state, movement, and Advanced Swarm Behavior telemetry. It is intended to support research in swarm intelligence, autonomous multi-agent systems, behavior analysis, network-aware coordination, machine learning, reinforcement learning, and multi-agent reinforcement learning. The dataset is simulation-generated and is not real-world flight telemetry; no benchmark, deployment, causal, safety, or operational-performance claim is made.

2. Dataset Identity and Motivation

The dataset provides a structured simulation resource for studying mission transitions, drone-level state evolution, movement dynamics, network conditions, coordination variables, and mission context. Its processed transition representation preserves explicit behavior telemetry alongside transition information, allowing researchers to formulate task-specific analyses and learning problems without treating the derived representations as a replacement for the rich source data.

Property	Verified value
Dataset	Golden Arrows Dataset V3.0
Version	3.0
Dataset type	Simulation-generated multi-drone swarm transition dataset
Missions	100
Transitions	11,570
Drone-transition records	347,100
Drones per transition	30
Mission types	PATROL, SURVEILLANCE, SEARCH, RECOVERY
Advanced Swarm Behavior	Enabled throughout the released records
Time step	0.5
Maximum steps	1000
Documented seed	20260916

Table 1. Dataset summary. Values are taken from the package documentation, metadata, schemas, validation artifacts, and direct file counts where applicable. The seed is documented as a configuration/reproducibility value; mission metadata also contains mission-specific seed values.

3. Dataset Composition

Mission type	Missions
PATROL	25
SURVEILLANCE	25
SEARCH	25
RECOVERY	25
Total	100

Table 2. Mission distribution. The four mission labels identify the mission families represented in the simulation; no comparative effectiveness or difficulty claim is made.

Raw mission records. The package preserves mission-level JSON records under `data/raw/missions/`. These records retain mission context and provenance information; they are not described as ground truth.

Processed rich transitions. The primary analytical representation is under `data/processed/transitions/`. Each JSONL object represents one transition and contains mission, transition, decision, action, result, network, and swarm information. The processed representation is the analytical source of truth for transition-level work.

4. Package Organization

```
Golden Arrows Dataset V3.0/
|-- README.md
|-- LICENSE
|-- CITATION.cff
|-- dataset_description.pdf
|-- dataset_card.md
|-- CHANGELOG.md
|
|-- data/
|   |-- metadata.csv
|   |-- raw/
|   |   |-- missions/
|   |   |-- processed/
|   |   |-- transitions/
|   |
|   |-- schema/
|   |   |-- transition_schema.json
|   |   |-- behavior_schema.json
|   |   |-- mission_schema.json
|   |
|   |-- validation/
|   |   |-- dataset_validation_report.txt
|   |   |-- dataset_statistics.json
|   |
|   |-- examples/preprocessing/
|   |   |-- README.md
|   |   |-- prepare_state_only.py
|   |   |-- prepare_behavior_aware.py
```

Figure 1. Final publication package organization. The package contains raw and processed data, schemas, validation artifacts, and optional preprocessing examples.

5. Simulation and Mission Context

The release represents a simulated swarm of 30 drones. PATROL missions represent patrol-oriented movement and coordination; SURVEILLANCE missions represent monitoring-oriented movement and coordination; SEARCH missions represent distributed search behavior; and RECOVERY missions represent recovery-oriented objectives and coordinated movement. These descriptions identify the configured mission contexts only; they do not rank mission types or establish real-world equivalence.

6. Data Generation and Reproducibility

The package documentation records a simulated generation configuration with 30 drones, 100 missions, 25 missions per type, `dt = 0.5`, `max_steps = 1000`, and documented seed 20260916. Reproducibility support includes mission metadata, raw mission JSON, processed transition JSONL, machine-readable schemas, validation artifacts, deterministic preprocessing input ordering, sorted mission IDs, deterministic row ordering, and mission-level splitting with split seed 20260916.

7. Conceptual System Architecture

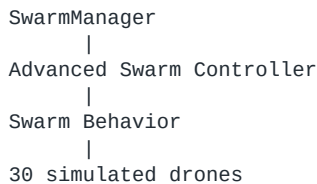


Figure 2. Conceptual architecture. SwarmManager is associated with mission/task assignment and role/task management. The Advanced Swarm Controller provides integration/orchestration between management and behavior. Swarm Behavior provides movement-related telemetry. This conceptual description does not assert undocumented implementation details.

The behavior system is scripted behavior telemetry. It is not a trained reinforcement-learning policy, a learned neural-network controller, an RL action generator, or evidence of trained-agent performance.

8. Transition Structure

The processed records are JSON Lines and follow the conceptual structure STATE -> ACTION -> RESULT -> NEXT STATE. At the top level, the verified record includes `schema_version`, `mission`, `transition`, `decision`, `action`, `result`, `network`, and `swarm`. The swarm object contains 30 drone records in each transition.

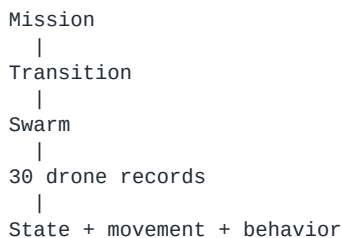


Figure 3. Transition-level organization. The machine-readable definitions are in the transition, behavior, and mission schema files.

9. Advanced Swarm Behavior

Advanced Swarm Behavior telemetry is preserved in each drone record. The seven recorded components are listed below with cautious descriptive wording.

Component	Description
Cohesion	Records a tendency associated with group cohesion.
Alignment	Records alignment-related movement behavior.
Separation	Records local separation behavior.
Formation	Records formation-related movement behavior.
Target Attraction	Records attraction toward the relevant mission target.
Network Preservation	Records network/connectivity-preservation behavior.
Boundary Avoidance	Records movement away from boundary-risk conditions.

Table 3. Advanced Swarm Behavior components. The table describes recorded telemetry and does not establish component independence, safety guarantees, or learned-policy behavior.

10. State-Only Preprocessing Example

`examples/preprocessing/prepare_state_only.py` is an optional executable example, not the authoritative dataset. It reads the rich processed transitions and writes local tabular artifacts when run by a researcher.

Category	Contents
Model inputs (13)	active_links; connected_drones; progress_before; position_x; position_y; altitude; speed; direction_x; direction_y; battery; signal_strength; latency; packet_loss
Metadata	mission_id; mission_type; drone_id; tick
Targets	next_position_x; next_position_y
Excluded fields	progress_after; actual_movement; behavior; action; decision

Table 4. State-only representation. position_x and position_y come from state_before.position. The target coordinates come from the recorded resulting state_after.position. Several physical and network inputs are also logged under state_after; the representation is not claimed to be a strict causal pre-action state.

11. Behavior-Aware Preprocessing Example

examples/preprocessing/prepare_behavior_aware.py is an optional executable example. It preserves the same base representation and adds verified Advanced Swarm Behavior telemetry.

Feature group	Count
Base state/context	13
Behavior-level telemetry	6
Behavior-vector features	8
Behavior-component features	28
Total model input features	55

Table 5. Behavior-aware feature composition.

Behavior-level features: speed_scale, neighbor_count, minimum_neighbor_distance, formation_error, connectivity_risk, and boundary_risk.

Behavior-vector features: behavior_current_position_x, behavior_current_position_y, behavior_current_velocity_x, behavior_current_velocity_y, behavior_desired_velocity_x, behavior_desired_velocity_y, behavior_movement_vector_x, and behavior_movement_vector_y.

Component	Recorded values
cohesion	x, y, weight, magnitude
alignment	x, y, weight, magnitude
separation	x, y, weight, magnitude
formation	x, y, weight, magnitude
target_attraction	x, y, weight, magnitude
network_preservation	x, y, weight, magnitude
boundary_avoidance	x, y, weight, magnitude

Table 6. Behavior-component fields: seven components times four values equals 28 features. Targets remain next_position_x and next_position_y. Excluded fields are actual_movement, action, decision, progress_after, and state_after.

Behavior telemetry is not an RL action.

12. Temporal Semantics

V3.0 contains fields recorded at different points in a transition. `state_before.position` provides the current position used by the documented derivatives. Several physical and network values are stored under `state_after`. Advanced Swarm Behavior telemetry is recorded separately from `actual_movement`. The targets `next_position_x` and `next_position_y` represent the resulting position recorded under `state_after.position`. Action, decision, progress, network, and behavior fields have their own transition semantics.

Fields stored under `state_after` must not automatically be interpreted as valid pre-action observations. Researchers should review temporal ordering for supervised learning, causal analysis, reinforcement learning, multi-agent reinforcement learning, forecasting, and control-policy learning. The examples do not solve all temporal leakage issues.

13. Data Leakage and Feature Interpretation

The preprocessing examples explicitly exclude documented outcome, control, and behavior fields according to their feature definitions and preserve metadata separately from model inputs. Feature exclusion does not guarantee elimination of all possible leakage. Researchers should independently review feature timing, target construction, mission structure, temporal dependencies, task formulation, identifiers, and transition semantics.

14. Mission-Level Splitting

Both examples use the documented deterministic mission-level split: 68 training missions, 16 validation missions, and 16 test missions, with 17/4/4 missions per type and split seed 20260916. Rows from one mission are temporally and structurally related; random row-level splitting could place related observations from the same mission into multiple partitions.

Split	Missions	Previously validated rows
Train	68	234,600
Validation	16	58,860
Test	16	53,640
Total	100	347,100

Table 7. Documented example split. The CSV derivatives are not included in the final publication package; these counts document the previously validated example split only. Unequal row counts arise because missions contain different numbers of transitions.

15. Running the Optional Examples

From `examples/preprocessing/`, use the published data directory as the source root:

```
python .\prepare_state_only.py --source-root ..\..\data
python .\prepare_behavior_aware.py --source-root ..\..\data
python .\prepare_state_only.py --help
python .\prepare_behavior_aware.py --help
```

The scripts also expose the implemented `--source-root`, `--output-root`, and `--split-seed` options. Generated CSVs, schemas, and reports are local artifacts and are not part of the released package.

16. Validation and Quality Control

The package audit established 100 raw mission files, 100 processed transition files, 100 missions, 11,570 transitions, 347,100 drone-transition records, 30 drones per transition, and 25 missions per type. Verified checks include schema consistency, transition integrity, drone identity consistency, behavior coverage, behavior-component coverage, duplicate detection, missing-value checks, numeric finiteness checks, mission-level split validation, and mission completion.

These artifacts assess structural and descriptive integrity. They do not prove scientific validity, real-world flight validity, safety certification, causal validity, operational effectiveness, or deployment readiness.

17. Validation Artifacts

The package contains `validation/dataset_validation_report.txt` and `validation/dataset_statistics.json`. These provide human-readable and machine-readable validation information supporting the released dataset. They are not evidence of model performance or scientific validity.

18. Intended Research Use

The dataset may support swarm behavior analysis, supervised learning, feature engineering, baseline machine-learning experiments, state-representation experiments, behavior-aware modeling, reinforcement-learning preprocessing, multi-agent reinforcement-learning preprocessing, network-aware swarm analysis, and mission-transition analysis. These are possible research uses; the package makes no guarantee about results or deployment suitability.

19. Limitations

- The dataset is simulation-generated and is not real-world flight telemetry.
- The mission controller is scripted.
- Advanced Swarm Behavior is scripted behavior telemetry, not a trained policy.
- Temporal semantics require task-specific review.
- Fields under `state_after` are not automatically pre-action observations.
- Feature exclusion does not guarantee elimination of all leakage.
- Researchers may need task-specific preprocessing.
- No deployment-readiness claim is made.
- No model-performance claim is made.
- No causal-validity guarantee is made.

20. Publication Scope

The final package intentionally publishes the authoritative rich transition data, raw mission records, metadata, schemas, validation artifacts, and optional preprocessing examples. It intentionally does not publish `train.csv`, `validation.csv`, `test.csv`, generated ML-ready datasets, `ml_ready` directories, analysis directories, development reports, pilot datasets, or staging directories.

```
Published authoritative dataset
|
data/raw/missions/
data/processed/transitions/
|
Optional researcher-run preprocessing
|
Locally generated ML representations
```

Figure 4. Publication scope. Derived ML outputs are generated locally by researchers and are not publication contents.

21. Final Package Structure

```
Golden_Arrows_Dataset_V3.0/  
|-- README.md  
|-- LICENSE  
|-- CITATION.cff  
|-- dataset_description.pdf  
|-- dataset_card.md  
|-- CHANGELOG.md  
|  
|-- data/  
|   |-- metadata.csv  
|   |-- raw/missions/  
|   |-- processed/transitions/  
|  
|-- schema/  
|   |-- transition_schema.json  
|   |-- behavior_schema.json  
|   |-- mission_schema.json  
|  
|-- validation/  
|   |-- dataset_validation_report.txt  
|   |-- dataset_statistics.json  
|  
|-- examples/preprocessing/  
|   |-- README.md  
|   |-- prepare_state_only.py  
|   |-- prepare_behavior_aware.py
```

22. Related Documentation and Citation

Related package documents are README.md, dataset_card.md, CITATION.cff, CHANGELOG.md, the schema files, the validation artifacts, and the preprocessing README. Citation and release metadata should be completed by the dataset maintainers when the version of record is registered; no author, affiliation, DOI, URL, or publication date is invented here.

23. Closing Statement

Golden Arrows Dataset V3.0 is a frozen simulation-generated resource for research and analysis of multi-drone swarm transitions. Its rich transition records and explicit behavior telemetry are intended to support transparent, task-specific research while preserving clear boundaries around temporal interpretation, simulation scope, and the absence of performance or deployment claims.